

Assignment – Week 3
Introduction to Machine Learning-IIT KGP

Topics: Instance Based Learning, Feature Selection, Feature Extraction,
Collaborative Filtering

TYPE OF QUESTION: MCQ/MSQ

Number of Questions: 10

Total Marks: $10 \times 2 = 20$

Question 1:

Suppose the following dataset is given where x_1 and x_2 are input variables and Class is the target variable.

x_1	x_2	Class
0	0	P
1	0	P
0	1	Q
2	1	P
2	2	Q
3	3	Q

Using Euclidean distance and 3-NN (K-NN with $k=3$), what will be the predicted class of the test point (1,1)?

- A. P
- B. Q
- C. Cannot determine
- D. Both classes equally likely

Correct Answer: A

Detailed Solution:

Compute the Euclidean distance from the test point (1,1) to all training points.

The nearest 3 neighbors are:

- (1,0) $\rightarrow$ P
- (0,1) $\rightarrow$ Q
- (2,1) $\rightarrow$ P

Among these nearest neighbors:

- P appears 2 times
- Q appears 1 time

Hence, the predicted class is: P

Question 2:

Suppose a classification problem contains 7 classes. What is the maximum number of discriminant vectors possible in LDA?

- A. 5
- B. 6
- C. 7
- D. 8

Correct Answer: B

Detailed Solution: LDA can produce at most $c-1$ discriminant vectors, where c is the number of classes. Therefore, $7-1 = 6$.

Question 3:

Which of the following statements about KNN are true?

- I. KNN stores the training data during training.
- II. KNN is a lazy learning algorithm.
- III. KNN requires distance computation during testing.

- A. I only
- B. II only
- C. I and II only
- D. I, II and III

Correct Answer: D

Detailed Solution: KNN memorizes the training data and computes distances during testing, making it a lazy learning algorithm.

Question 4:

Which of the following is an example of feature extraction?

- A. Removing highly correlated features
- B. Selecting the top 5 important features
- C. Converting images into principal components using PCA
- D. Deleting noisy samples

Correct Answer: C

Detailed Solution: PCA transforms original features into new principal components and is therefore a feature extraction method.

Question 5:

Feature selection is useful because:

- A. It reduces redundant features
- B. It may reduce overfitting
- C. It can reduce computational cost
- D. All of the above

Correct Answer: D

Detailed Solution: Feature selection removes irrelevant or redundant features, reduces computational cost and may improve generalization.

Question 6:

When the value of k in KNN is very small, the model is more likely to:

- A. Underfit
- B. Overfit
- C. Become linear
- D. Reduce variance completely

Correct Answer: B

Detailed Solution: Small values of k make KNN highly sensitive to noise, increasing overfitting.

Question 7:

Which of the following statements about PCA are true?

- I. PCA is an unsupervised method.
- II. PCA reduces dimensionality.
- III. PCA maximizes variance along principal components.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II and III

Correct Answer: D

Detailed Solution: All three statements are correct.

Question 8:

Which of the following describes a key algorithmic difference between Linear Discriminant Analysis (LDA) and Principal Component Analysis (PCA)?

- A. PCA maximizes the total variance of the data projection, while LDA maximizes class separability by utilizing class labels.
- B. LDA is entirely unsupervised, whereas PCA requires explicit class labels to construct principal components.
- C. PCA reduces features by selecting a subset of the original features, while LDA always constructs completely new features.
- D. LDA can produce an infinite number of discriminant vectors regardless of the number of target classes.

Correct Answer: A.

Detailed Solution: PCA is an unsupervised feature extraction method that ignores class labels and finds orthogonal axes (principal components) that maximize the total variance of the data. LDA is a supervised feature extraction method that explicitly uses class labels to maximize the between-class variance while minimizing the within-class variance, ensuring optimal class separability.

Question 9:

Identify whether the following statement is true or false:

“Feature extraction creates new transformed features from existing features.”

- A. TRUE
- B. FALSE

Correct Answer: A

Detailed Solution: Feature extraction transforms existing features into new lower-dimensional representations.

Question 10:

Find the value of the Pearson correlation coefficient between X and Y for the following data:

X	Y
1	2
2	4
3	6
4	8

- A. 0
- B. 0.5
- C. 1
- D. -1

Correct Answer: C

Detailed Solution: Since $Y = 2X$, there is a perfect positive linear relationship between X and Y. Therefore the Pearson correlation coefficient is 1.
